# Arjun Nanduri

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## Education

**University of California: Berkeley**, BA in Computer Science — Expected May 2028

- GPA: 3.9 | SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Computer Security, Data Structures

## Research

**Data Science Research Intern**, CDSS, UC Berkeley – Berkeley, CA — Jan 2025 – May 2025

- Fine-tuned BART transformer for Ancient Akkadian lemmatization in low-resource settings, **beating open-source baselines by 15%** using Levenshtein distance over lemma forms.
- Built preprocessing and alignment pipelines for structured linguistic data from ORACC's cuneiform corpus ($\sim$**1M annotated words across** $\sim$**10K texts**).
- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.

## Experience

**Machine Learning Engineer**, Aver Technologies Inc. – Virginia — Aug 2025 – Present

- Building a multi-stage PyTorch CV pipeline to automate depth tracking when driving piles (foundational support for heavy structures, like bridges) into the ground — **replacing a manual, hand-marked DOT compliance process**.
- Trained a CV pipeline for **real-time pile depth tracking** — object detection to localize the pile, YOLO segmentation and line detection to track painted depth markings, an open-source digit detector to identify line numbers, and audio-visual synchronization to isolate each hammer strike, converting pixel-space line displacement into inch/foot measurements via a depth map.
- Conducted field interviews with construction workers to surface real-world constraints — inconsistent hand-drawn markings, Spanish-speaking workforce — **directly informing model design decisions**.
- Field interviews also surfaced that a LIDAR-based measurement approach was cost-prohibitive given the project's thin margins, and that workers frequently crossed the camera's line of sight during recording — **informed the decision to pursue a CV-based approach over LIDAR** for cost-effectiveness.
- **Helped file a provisional patent**, designing a tripod-mounted 1000fps camera and Jetson Orin GPU capture rig.
- Designing two-phase roadmap: Phase 1 CV digitization of manual process, **Phase 2 LIDAR elimination of markings**.

## Projects

**Quadruped Locomotion with PPO** (MuJoCo, Gymnasium, SB3) — github.com/FavouritePlayer/ant_sim

- Trained PPO quadruped policies in MuJoCo/Gymnasium, benchmarking flat-ground baselines against terrain-adapted and damage-robust variants across **10 matched random seeds per condition**.
- Terrain adaptation: policy trained on randomized heightfields **achieves 2.1x the reward** (893 vs. 424) and 40% lower fall rate than the flat-trained baseline on unseen rough terrain.
- Leg-damage robustness: a specialist policy trained with one leg permanently amputated reaches $\sim$49x the flat-baseline reward on that leg, but **does not transfer to other legs** (100% fall rate).
- Built a compound policy router that switches checkpoints by damaged leg, **beating any single policy** (836 vs. 497 mean reward) across a 4-leg $\times$ 10-seed evaluation sweep.
- Validated all headline results via **multi-seed training replication and an automated regression test suite**; generated comparison videos for each experiment.

**Face Morpher — From-Scratch Warping** (NumPy, dlib, scipy) — github.com/FavouritePlayer/face-morpher

- Implemented affine warping, inverse mapping, and bilinear interpolation **from scratch in NumPy** to remap pixels between corresponding faces.
- Solved a per-triangle affine transform from 3-point landmark correspondences (**3-point** $\rightarrow$ **2$\times$3 matrix**), combined with polygon-masked, piecewise Delaunay-triangle warping to compute image morphs and mean faces.

## Skills

**Languages:** Python, SQL

**Technologies:** PyTorch, MuJoCo, Gymnasium, Stable-Baselines3, NumPy, dlib, scipy, Git